

Automated Information Extraction Pipeline for Constructing ED Knowledge Graphs from Korean Clinical Problem Lists

Hyeyoon Moon^a, Eunhye Jang^a, Won Chul Cha^{a, b}

^a Department of Digital Health, SAIHST, Sungkyunkwan University, Seoul, Korea ^b Department of Emergency Medicine, Samsung Medical Center, Seoul, Korea

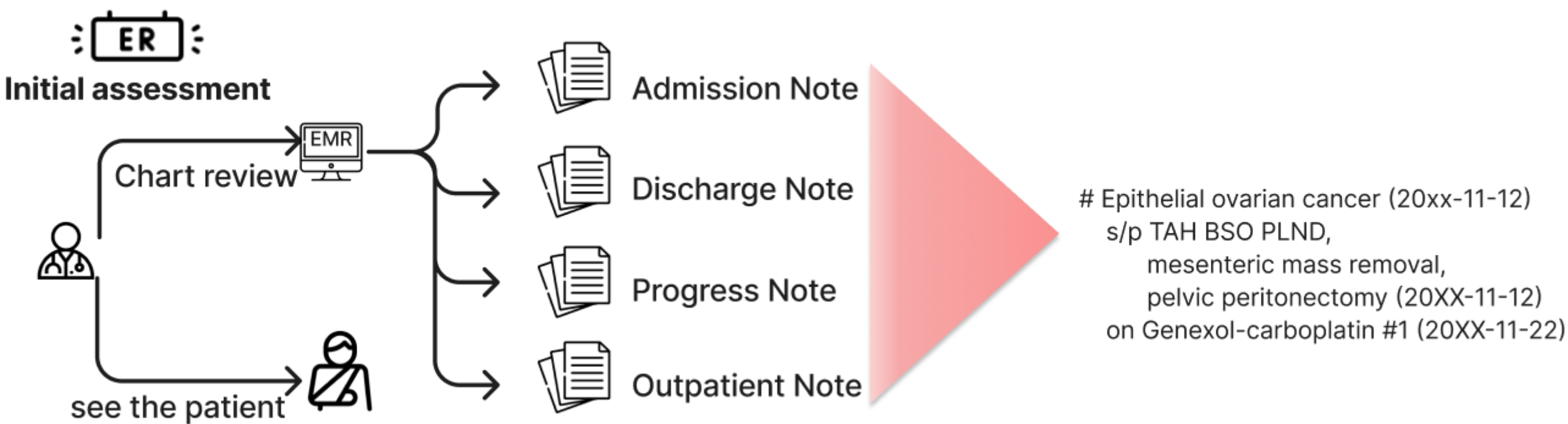
KEY FINDING More standardization is not better.

Partial normalization (NED) outperforms both raw text (+4.48% F1) and full SNOMED CT standardization (−5.97%). Critical clinical details(laterality, staging, procedural history) are preserved in NED but collapsed by FSN, making partial normalization the safer choice for downstream clinical AI.

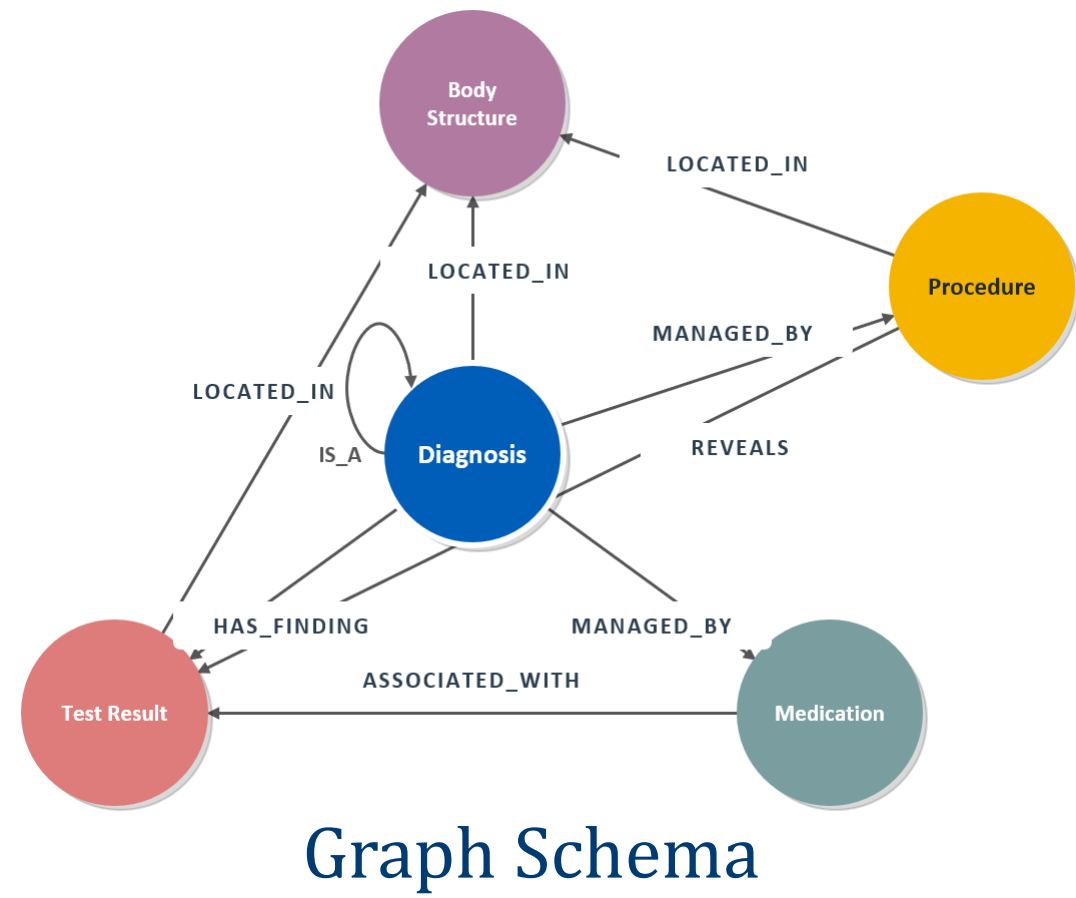
1 Background

In tertiary ED settings, each patient generates **heterogeneous Korean–English clinical notes** (admission, discharge, progress, outpatient) where diagnoses are progressively refined across time. Capturing this longitudinal progression requires **automated relation extraction** beyond entity-level NER — linking diverse expressions of the same condition into a coherent representation.

Integrating this disease progression and terminological diversity into a coherent representation requires automated relation extraction across heterogeneous notes.



Goal: Construct a structured ED knowledge graph by automatically extracting entities and clinical relations from heterogeneous notes — unifying diverse expressions and longitudinal disease progression into a queryable representation that supports decision-making and downstream clinical AI.

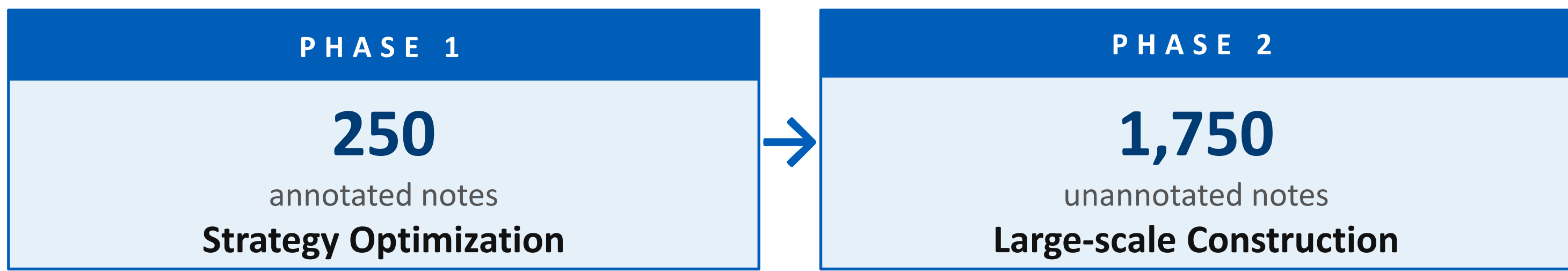


2 Research Questions

- RQ1** What level of input normalization optimizes LLM performance in relation extraction?
- RQ2** Which model family — general LLM vs reasoning-trained — performs best for this task?

3 Dataset & Two-Phase Strategy

Source: Samsung Medical Center ED clinical notes (NER pre-tagged, IRB-approved)



→ Maximizes KG scale while minimizing annotation cost

4 Normalization Strategies

STRATEGY A Raw Text	STRATEGY B NED BEST	STRATEGY C SNOMED CT FSN
DATA Korean–English, abbreviations mixed clinical text as written	DATA Expanded English full terms via Named Entity Disambiguation(NED)	DATA Standardized SNOMED CT concepts
STRATEGY No transformation	STRATEGY Context-aware expansion via Gemma 3 12B	STRATEGY MedCAT + SapBERT mapping
EXAMPLE Stomach ca, cT3N1	EXAMPLE Stomach cancer, clinical T3N1	EXAMPLE Malignant tumor of stomach (SNOMED CT 363349007)

FSN collapses TNM staging and c/p distinction (both critical for treatment decisions.)

5 Relation Extraction Pipeline

- 1 Few-shot Prompting**
4 examples covering all 5 relation types and diverse entity-type combinations, including common ED patterns (s/p, due to, r/o)
- 2 LLM Inference**
Generates relation triplets in JSON format
- 3 Schema-based Validation**
Filters triplets violating entity-pair rules (e.g., Diagnosis–Diagnosis allows only IS_A / ASSOCIATED_WITH)

6 Results

Strategy	F1	Δ vs Raw
Raw Text	0.67	baseline
NED	0.70	+4.48%
SNOMED CT FSN	0.63	−5.97%

Table 1. Normalization Effect

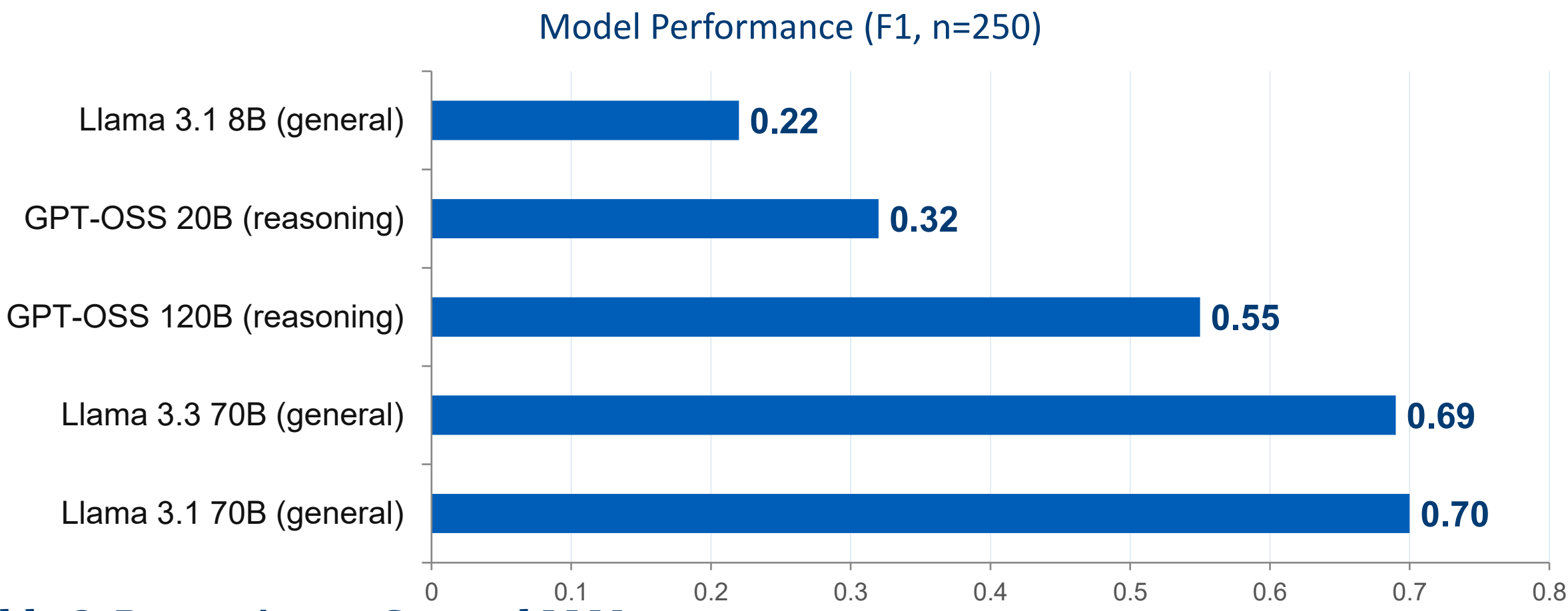


Table 2. Reasoning vs General LLM

INSIGHT 1 — Normalization	INSIGHT 2 — Model Family
Partial > Full. NED (+4.48%) beats both Raw and FSN. Full standardization over-collapses <i>clinical granularity</i> (FSN: −5.97%).	General > Reasoning. Llama 70B beats GPT-OSS 120B by 0.15 F1. RE is a pattern-matching task — long CoT reasoning offers no advantage.

7 Final Knowledge Graph

1,983 ENTITIES	3,208 RELATIONS
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from 2,000 ED problem lists · 4 node types · 5 relation types

MANAGED_BY	63.0%
ASSOCIATED_WITH	20.8%
Others	16.2%

CONCLUSION

Preserving clinical granularity is the key. standardization enables shared representations, but over-standardization erases the nuance clinicians and downstream AI depend on. NED balances both — a principle transferable to multilingual clinical NLP beyond Korean ED.

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Link to full paper & contact info

